

Where the Forest Goes Dark: Halo-centered Characterization of High Column Density Systems with IllustrisTNG

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ABSTRACT

High column density systems of neutral hydrogen (HCDs) — Lyman limit systems (LLSs), sub-damped and damped Ly α absorbers (DLAs) — contaminate both the 3D and the 1D statistics of the Ly α forest. Most DLAs are masked out from the analyses, but cleaning algorithms are not perfect, and weaker LLSs are individually undetectable, so the residual contamination is absorbed into nuisance parameters that dilute the cosmological constraining power. Simulations provide the ideal setting to study this effect, allowing absorption features to be directly traced back to the gas producing them. We identify high-column-density structures on the gas cells of the TNG50 hydrodynamical simulation, deblend them in velocity space, and split every sightline into HCD-only and forest-only spectra. Applied to halos at $z \simeq 3$ out to 50 virial radii, it yields each class's covering fraction against impact parameter b and halo mass. The impact parameter defines the type of absorption appearing in the spectrum: DLAs give way to sub-DLAs at $b \approx 0.11 R_{200c}$, sub-DLAs to LLSs at 0.3, and LLSs to the Ly α -forest at 0.5, which covers $\simeq 78\%$ of sightlines at R_{200c} . In units of R_{200c} the sequence is nearly mass-independent over three decades in mass, so the virial radius sets the scale of the neutral gas distribution. A single Voigt component recovers the column density of essentially every damped system, but not of LLSs, whose absorption features often arise from several separate contributions from gas structures along the sightline. Our detailed characterization of HCDs gives the first step to the construction of advanced techniques to directly forward-model their contribution to Ly α spectra.

Keywords: Lyman alpha forest (980) — Quasar absorption line spectroscopy (1317) — Damped Lyman-alpha systems (349) — Lyman limit systems (981) — Intergalactic medium (813) — Circumgalactic medium (1879) — Hydrodynamical simulations (767)

1. INTRODUCTION

The Ly α forest — the dense set of absorption features imprinted on the spectra of high-redshift quasars (QSOs) by intergalactic neutral hydrogen (for reviews, see M. Rauch 1998; A. A. Meiksin 2009; M. McQuinn 2016) — is one of the few probes of structure formation available at $2 \lesssim z \lesssim 5$. Because the transmitted flux traces the underlying density field (e.g., R. Cen et al. 1994; L. Hernquist et al. 1996; R. A. C. Croft et al. 1998), its statistics constrain the matter power spectrum

on quasi-linear and non-linear scales (e.g., P. McDonald et al. 2006), the thermal and ionization state of the intergalactic medium (IGM; L. Hui & N. Y. Gnedin 1997; J. Schaye et al. 2000; M. Walther et al. 2019), and the expansion history through baryon acoustic oscillations (BAO; N. G. Busca et al. 2013; J. E. Bautista et al. 2017; H. du Mas des Bourboux et al. 2020).

The number of available forests has grown by more than two orders of magnitude in two decades: from samples of order 10^3 spectra in the early 2000s (P. McDonald et al. 2006) to $\simeq 820\,000$ in the second data release of the Dark Energy Spectroscopic Instrument (DESI; DESI Collaboration et al. 2016, 2022; M. Abdul Karim et al. 2025). These data constrain the isotropic BAO

scale at $z_{\text{eff}} = 2.33$ to 0.65% and the line-of-sight scale, which fixes $H(z)$, to 1.1% (M. Abdul Karim et al. 2025), and support full-shape analyses of the three-dimensional correlations that extract the Alcock–Paczynski signal in addition to the acoustic scale (DESI Collaboration et al. 2026). The first data release already yielded measurements of the one-dimensional flux power spectrum (P_{1D}) with two independent estimators (N. G. Karaçaylı et al. 2025; C. Ravoux et al. 2025), extending the BOSS and eBOSS sequence (N. Palanque-Delabrouille et al. 2013; S. Chabanier et al. 2019; C. Ravoux et al. 2023) and constraining the amplitude and logarithmic slope of the linear matter power spectrum at $z = 3$ (J. Chaves-Montero et al. 2026).

Statistical uncertainties play a central role in these measurements and the treatment of astrophysical contaminants costs a substantial part of the constraining power. Among the leading contaminants are high column density systems (HCDs): regions of neutral hydrogen dense enough to be optically thick, encountered when a sightline passes through the circumgalactic medium (CGM) of a galaxy or other dense gas. The absorption features of these systems saturate the transmitted flux over several Å of spectrum, where the forest features it hides are themselves a fraction of an Å wide. The affected range grows with column density. HCDs are classified by their H I column density, N_{HI} , into Lyman limit systems (LLSs), sub-damped Ly α systems (sub-DLAs) and damped Ly α systems (DLAs), in order of increasing self-shielding.⁴

The resulting contamination of the forest observables is scale dependent, and has been measured. A. Font-Ribera & J. Miralda-Escudé (2012) showed analytically (and also employing mock spectra) that unmasked HCDs raise the effective Ly α bias and lower the redshift-space distortion parameter, and M. McQuinn & M. White (2011) estimated the DLA correlations between sightlines. K. K. Rogers et al. (2018a) and K. K. Rogers et al. (2018b) measured the effect in cosmological hydrodynamical simulations, provided multiplicative P_{1D} templates as a function of N_{HI} , $k_{\parallel}$ and redshift, and found that residual contamination of the three-dimensional flux power spectrum reaches the ten-

percent level at $k \sim 0.1 h \text{ Mpc}^{-1}$ even after the strongest systems are masked.

Survey analyses adopt these prescriptions. BAO and full-shape fits marginalize over an HCD bias and a wing-scale parameter (H. du Mas des Bourboux et al. 2020; A. Adame et al. 2025). The DESI DR1 P_{1D} analysis splits the population into column-density groups following K. K. Rogers et al. (2018a), because DLA finders lose completeness rapidly below the damped threshold (J. Chaves-Montero et al. 2026). The amplitude of each group is left free during inference, but that freedom is expensive: The LLS amplitudes rank among the nuisance parameters most strongly correlated with the recovered cosmology, marginalizing over them inflates the uncertainty on the compressed parameters by $\simeq 50\%$, and dropping the LLS and sub-DLA terms shifts the inferred parameters by more than 3σ (J. Chaves-Montero et al. 2026). T. Tan et al. (2025) tested a Voigt-based alternative against mocks with known HCD content and found it comparable to the templates in use.

What all of these descriptions have in common is that they parametrize HCDs by their column density and by an effective large-scale bias, calibrated either on the observed column-density distribution function or on the measured DLA–forest cross-correlation (A. Font-Ribera et al. 2012; I. Pérez-Ràfols et al. 2018). The link to the halos that host the absorbers has been studied mostly through this integrated bias, or through the abundance and metallicity of the DLA population (S. Bird et al. 2014), rather than through the spectral signature that a sightline actually records as a function of where it passes relative to a halo. Observations cannot supply this description. A spectrum records the absorption but not the gas that produced it, so a feature can be attributed to a halo of known mass at a known transverse distance only in simulations. A forward model that places HCDs on a simulated density or halo field, rather than drawing them from a global distribution, requires exactly that attribution, and we measure it here. D. Muñoz Santos et al. (2025) takes the complementary route in TNG50, starting from strong blended Ly α absorbers, at $\log N_{\text{HI}} \simeq 16$ and therefore below the LLS threshold, and identifying the halos they trace; we start from the halos and measure what each sightline records as a function of its distance from them.

HCDs are also of interest in their own right. Their incidence, kinematics and line profiles depend on the state of the CGM gas and the dense IGM, and therefore on self-shielding, accretion and feedback (J. Schaye 2001; A. Rahmati et al. 2013; C. Peroux & D. Nelson 2024). A characterization organized by halo properties is thus useful on both fronts.

⁴ Class boundaries in $\log_{10}(N_{\text{HI}}/\text{cm}^{-2})$, in order of increasing column density: the diffuse forest below 17.2; LLSs between 17.2 and 19, optically thick to ionizing radiation; sub-DLAs between 19 and 20.3, partially self-shielded; and DLAs above 20.3, essentially fully self-shielded (A. M. Wolfe et al. 2005). We split the damped regime further at 21.0 into small and large DLAs, since the extent of the contamination varies substantially across that range.

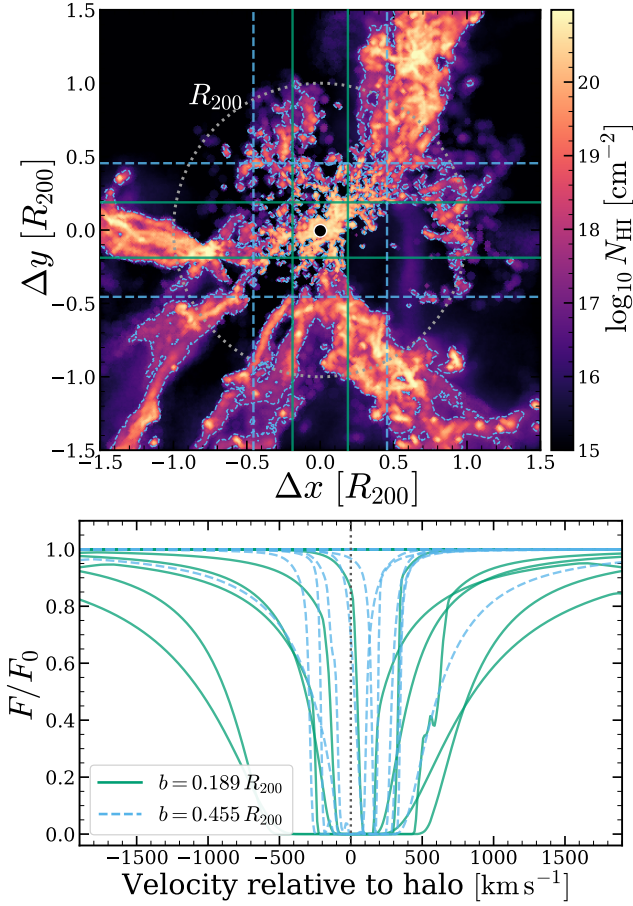


Figure 1. The halo-centered sampling of this work, for a halo of $M_{200c} \approx 10^{11.86} M_{\odot}$ at $z \simeq 3$. **Top:** H I column density projected along the z axis, with the halo centre marked by the black dot and R_{200c} by the grey dotted circle. The blue dashed contour traces $N_{\text{HI}} = 10^{17.2} \text{ cm}^{-2}$, the HCD threshold of Sect. 2.3. Straight lines mark the sightlines at two impact parameters, $b = 0.189 R_{200c}$ (green, solid) and $b = 0.455 R_{200c}$ (blue, dashed); those along x appear as horizontal lines and those along y as vertical lines, while the four along z are seen end-on and are not marked. **Bottom:** HCD-only transmitted flux recorded by the twelve sightlines at each impact parameter, including those along z , against velocity relative to the systemic velocity of the halo, with the same colour and line-style coding.

In this work we measure the imprint of HCDs on synthetic Ly α skewers crossing the CGM and near environment of halos at $z \simeq 3$. Figure 1 illustrates the approach on a single halo: sightlines are placed at controlled distances from a halo of known mass, and the absorption each one records is measured as a function of that distance.

Our measurements rest on three ingredients. First, we extend the `fake_spectra` package (S. Bird 2017) with a skewer-level identification procedure that flags the gas cells belonging to HCD structures before computing the

optical depth, separates structures whose absorption an observer could resolve into distinct systems, and returns two complementary spectra per sightline: `HCD_only`, carrying the flagged structures, and `Forest_only`, carrying the absorption that remains once they are removed.⁵ Second, we design a halo-centered sampling campaign that pierces individual haloes along a controlled grid of impact parameters, so that the resulting absorption statistics can be read directly as functions of halo mass and halo-centric distance. Third, we run the entire campaign on three levels of the TNG50 resolution ladder (R. Weinberger et al. 2017; V. Springel et al. 2018; A. Pillepich et al. 2018a; J. P. Naiman et al. 2018; A. Pillepich et al. 2018b; D. Nelson et al. 2019), which share a single baryonic model and therefore isolate numerical resolution as a systematic to be quantified rather than as a physical effect.

These three ingredients yield a systematic characterization of HCD absorption as a function of halo properties: how the absorber population is organized radially around halos, how that organization depends on host-halo mass, and how far a single-absorber description holds for the systems the pipeline isolates.

The paper is organized as follows. Section 2 describes the simulations, the skewer-extraction pipeline and the HCD isolation procedure, and defines the halo-centered sampling campaign. Section 3 presents the measurements: the radial organization of the absorber population, its dependence on host-halo mass and the skewer-level column-density distributions (Section 3.1), and the Voigt-profile characterization (Section 3.2). Section 4 summarizes our conclusions and their implications for forest modeling. Appendix A establishes the numerical convergence of every observable used in the main text and defines the range of impact parameters over which we regard our measurements reliable.

Throughout, we adopt the cosmology of the IllustrisTNG simulations, $\Omega_m = 0.3089$, $\Omega_b = 0.0486$, $\Omega_{\Lambda} = 0.6911$, $h = 0.6774$, $\sigma_8 = 0.8159$ and $n_s = 0.9667$ (Planck Collaboration et al. 2016).

2. METHODS

In the following subsections we explain how this work generates synthetic Ly α spectra from cosmological hydrodynamical simulations and isolate HCD features. Working with simulations grants direct access to the gas distribution behind every absorption feature, so each absorption can be assigned to the structure that produces it. This section describes the simulations (Sect. 2.1), the extraction of the skewers and the construction of

⁵ https://github.com/VictorRoberto132/FakeSpectra_DLAs